

## THE QUESTION

Q.13 · DMG 2019

**In ocean, regional strike-slip faults associated with lithospheric plate boundaries are known as**

**a Transform faults**

b Tear faults

c Wrench faults

d Scissor faults

CORRECT ANSWER

A

**Transform faults — plate boundaries where lithosphere is neither created nor destroyed**

## TEXTBOOK DEFINITIONS — THE AUTHORITY

TABLE 8.2 — TYPES OF FAULTS · VAN DER PLUIJM &amp; MARSHAK · PAGE 171

**Transform fault:** "In the preferred sense, transform faults are **plate boundaries at which lithosphere is neither created nor destroyed**. In a general sense, a transform fault links two other faults and accommodates the relative motion between the blocks of rock that move because of the displacement on the other two faults."

📖 Table 8.2, Page 171 — Chapter 8

TABLE 19.1 — TERMINOLOGY FOR STRIKE-SLIP FAULTS · PAGE 478

**Transform fault:** "A strike-slip fault that has the following characteristics: once formed, displacement across it can be constant along the length of the fault; displacement across it can be much greater than the length of the active fault; its length can be constant, increase, or decrease with time; it terminates at another fault. **Transform faults can be lithosphere plate boundaries**, in which case they terminate at intersections with other plate boundaries."

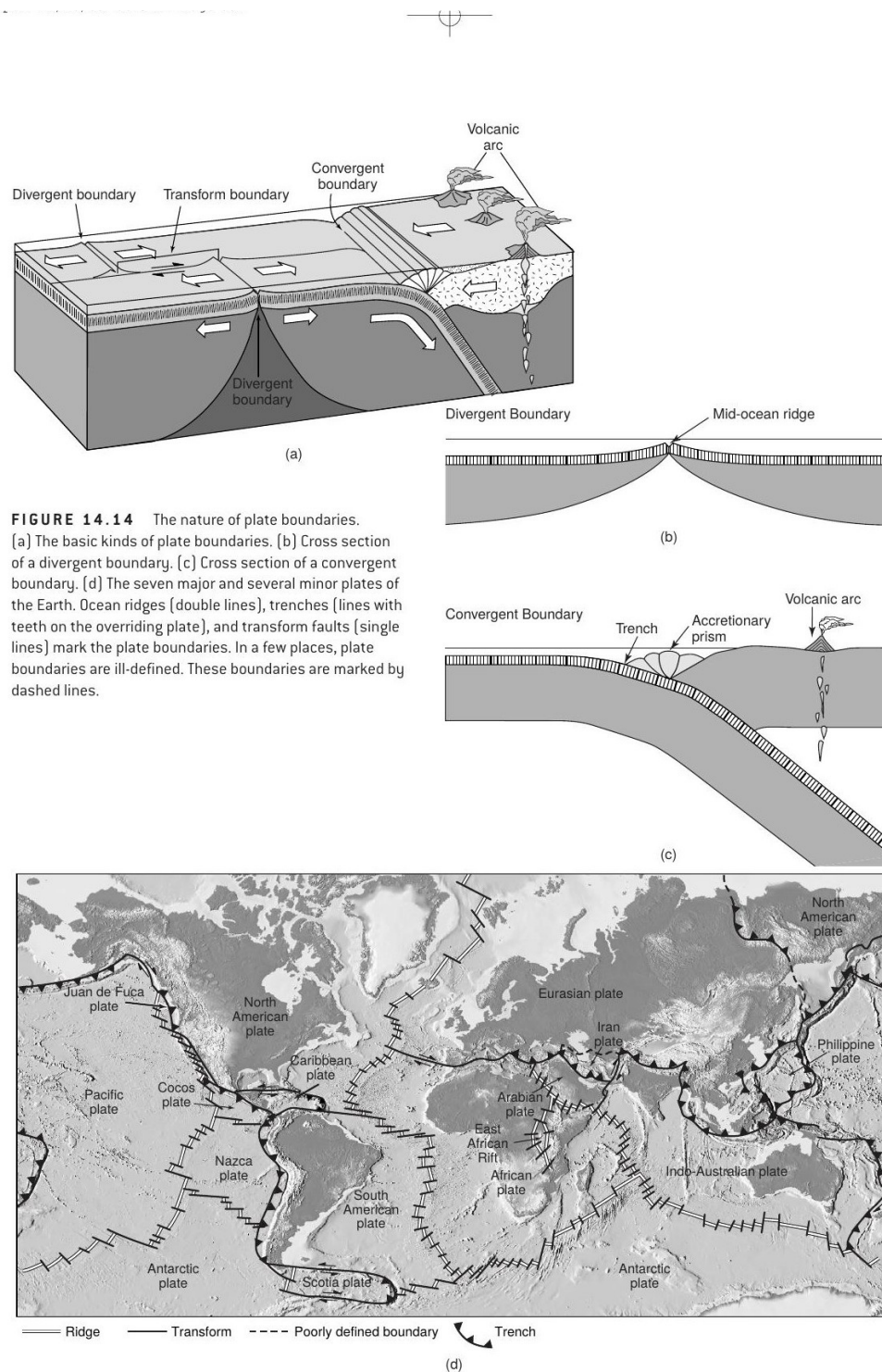
📖 Table 19.1, Page 478 — Chapter 19: Strike-Slip Tectonics

§19.2.1 TRANSFORM FAULTS · PAGE 479

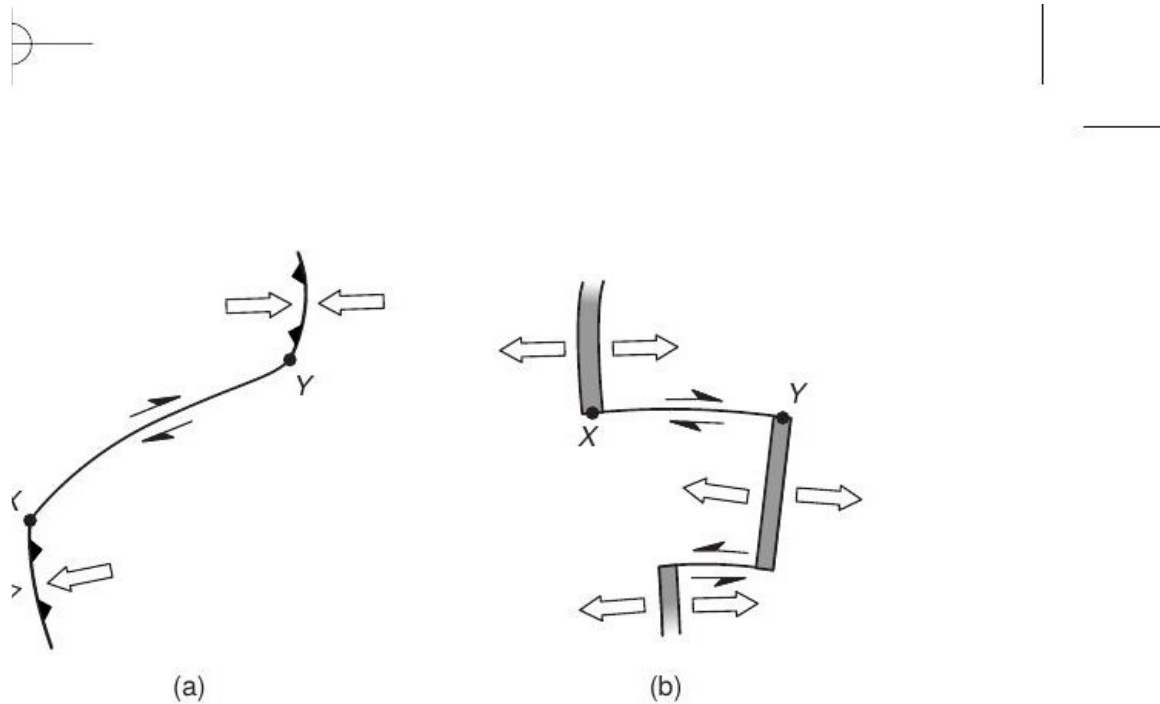
"J. Tuzo Wilson introduced the term 'transform faults' to the geologic literature in the early 1960s, in reference to **the third category of plate boundary** (distinct from convergent and divergent boundaries) ... Strike-slip faults in oceanic crust most commonly occur along **mid-ocean ridges**, where they trend perpendicular to the ridge axis and link segments of the ridge."

"At **transform plate boundaries**, one plate slides past another along a strike-slip fault. Since no new plate is created and no old plate is consumed along a transform, such a boundary can also be called a 'conservative boundary.' Transform plate boundaries can occur either in **continental or oceanic lithosphere**."

## FIGURES FROM THE TEXTBOOK

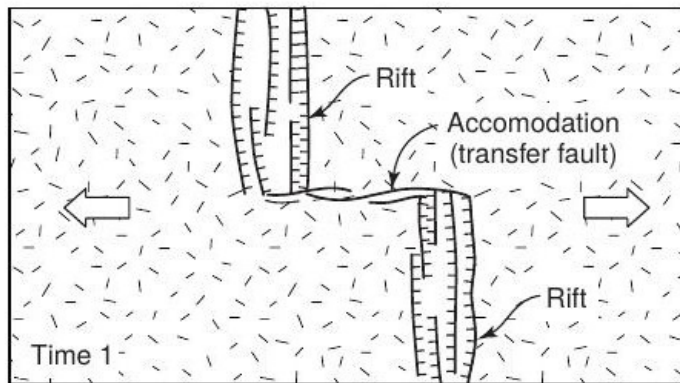


**FIGURE 14.14** — The nature of plate boundaries. (a) The three types: divergent boundary (mid-ocean ridge), convergent

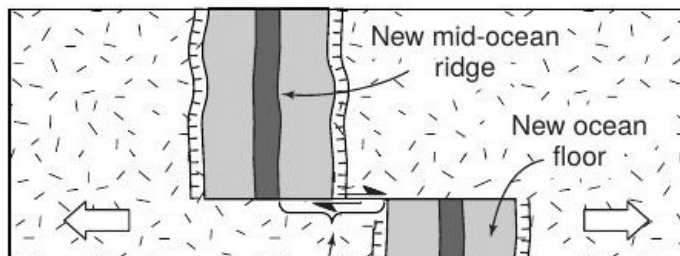


**FIGURE 19.3** (a) Map sketch showing a transform fault linking convergent plate boundaries. The fault trace occurs only between X and Y. (b) Map sketch showing a transform fault linking two ridge segments.

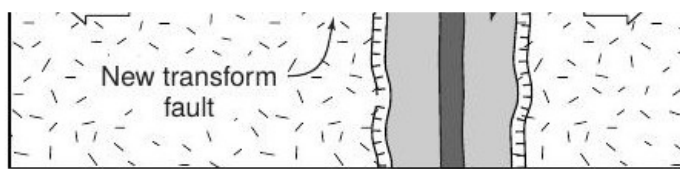
**FIGURE 19.3** — (a) Transform fault linking two convergent plate boundaries. (b) Transform fault linking two ridge segments. The active fault trace exists only between X and Y.



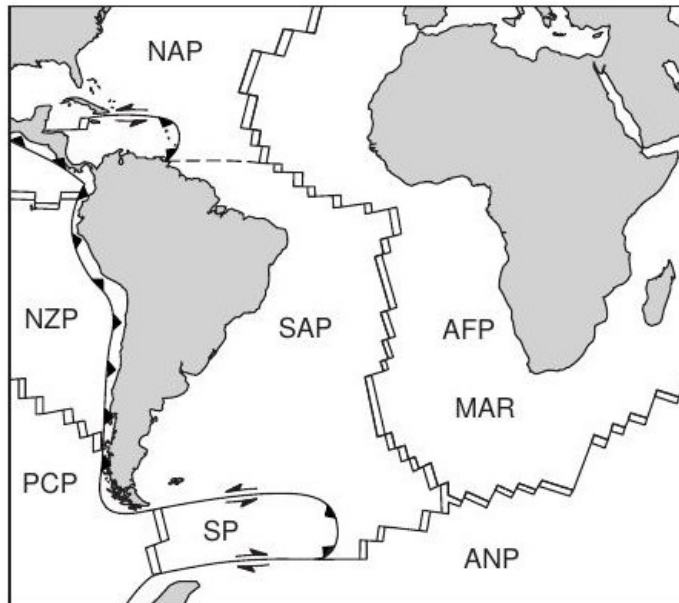
(a)



**FIGURE 19.7a-b** — (a) Transfer fault linking two rift segments. (b) As rifting succeeds, a new mid-ocean ridge forms and the transfer fault **evolves into a classic oceanic transform fault**.



(b)



(c)

**FIGURE 19.7** [a] Sketch map showing a transfer fault linking two rift segments. [b] Later, the rift segments have evolved into new mid-ocean ridges, and the transfer fault has evolved into an oceanic transform fault. Note that this fault still has the same length as when it originated. [c] Simplified map showing the transform faults along the Mid-Atlantic Ridge in the South Atlantic Ocean. SAP = South American Plate, NAP = North American Plate, AFP = African Plate, ANP = Antarctic Plate, NZP = Nasca Plate, CS = Caribbean Sea, SS = Scotia Sea, MAR = Mid-Atlantic Ridge.

## 9.2 TRANSFORM VERSUS TRANSCURRENT FAULTS 481

**FIGURE 19.7c** — Simplified map showing transform faults along the Mid-Atlantic Ridge in the South Atlantic Ocean, linking segments of the ridge. SAP = South American Plate, AFP = African Plate, NAP = North American Plate, ANP = Antarctic Plate, MAR = Mid-Atlantic Ridge. These are classic examples of **oceanic regional strike-slip faults at lithospheric plate boundaries** — i.e., **transform faults**.

## ★ J. TUZO WILSON — THE FATHER OF TRANSFORM FAULTS (1960S)

J. Tuzo Wilson coined the term "**transform faults**" to describe the third type of plate boundary — distinct from divergent (mid-ocean ridges) and convergent (subduction zones). He recognized that in ocean basins, mid-ocean ridges are not continuous features but are offset by **regional strike-slip faults** that link ridge segments. These faults are plate boundaries where plates simply *slide past* each other — no new crust is created, and no old crust is destroyed. This makes them "**conservative boundaries**."

The table below (directly from Van der Pluijm & Marshak, Table 8.2, p. 171) gives all fault types — reproduced here for reference:

| TABLE 8.2           | TYPES OF FAULTS                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Allochthon          | The thrust sheet above a detachment is the allochthon (meaning that it is composed of <i>allochthonous</i> rock; i.e., rock that has moved substantially from its place of origin).                                                                                                                                                                                                                                              |
| Autochthon          | The footwall below a detachment is the autochthon; it is composed of <i>autochthonous</i> rock, or rock that is still in its place of origin.                                                                                                                                                                                                                                                                                    |
| Contractional fault | A contractional fault is one whose displacement results in shortening of the layers that the fault cuts, regardless of the orientation of the fault with respect to horizontal.                                                                                                                                                                                                                                                  |
| Décollement         | The French word for detachment.                                                                                                                                                                                                                                                                                                                                                                                                  |
| Detachment fault    | This term is used for faults that initiate as a horizontal or subhorizontal surface along which the hanging-wall sheet of rock moved relative to the footwall. An older term "overthrust" is a regional detachment fault on which there has been a thrust sense of movement. Some detachments are listric, and on some detachments, regional normal-sense displacement occurs.                                                   |
| Dip-slip fault      | The slip direction on a dip-slip fault is approximately parallel to the dip of the fault (i.e., has a rake between $\sim 80^\circ$ and $90^\circ$ ).                                                                                                                                                                                                                                                                             |
| Extensional fault   | An extensional fault is one whose displacement results in extension of the layers that the fault cuts, regardless of the orientation of the fault with respect to horizontal.                                                                                                                                                                                                                                                    |
| Normal fault        | A normal fault is a dip-slip fault on which the hanging wall has slipped down relative to the footwall.                                                                                                                                                                                                                                                                                                                          |
| Oblique-slip fault  | The slip direction on an oblique-slip fault has a rake that is not parallel to the strike or dip of the fault. In the field, faults with a slip direction between $\sim 10^\circ$ and $\sim 80^\circ$ are generally called oblique-slip.                                                                                                                                                                                         |
| Overthrust fault    | This is an older term that you may find in older papers on faults, but is no longer used much today. The term is used for thrust faults of regional extent. In this context, "regional extent" means that the thrust sheet has an area measured in tens to hundreds of square km, and the amount of slip on the fault is measured in km or tens of km. Today, such faults are generally called regional detachments.             |
| Par-autochthonous   | If a fault block has only moved a small distance from its original position, the sheet is par-autochthonous (literally, relatively in place).                                                                                                                                                                                                                                                                                    |
| Reverse fault       | A reverse fault is a dip-slip fault on which the hanging wall has slipped up relative to the footwall.                                                                                                                                                                                                                                                                                                                           |
| Scissors fault      | On a scissors fault, the amount of slip changes along strike so that the hanging-wall block rotates around an axis that is perpendicular to the fault surface (Figure 8.4i).                                                                                                                                                                                                                                                     |
| Strike-slip fault   | The slip direction on a strike-slip fault is approximately parallel to the fault strike (i.e., the line representing slip direction has a rake [pitch] in the fault plane of less than $\sim 10^\circ$ ). Strike-slip faults are generally steeply dipping to vertical.                                                                                                                                                          |
| Transfer fault      | A transfer fault accommodates the relative motion between blocks of rock that move because of the displacement on other faults.                                                                                                                                                                                                                                                                                                  |
| Transform fault     | In the preferred sense, transform faults are plate boundaries at which lithosphere is neither created nor destroyed. In a general sense, a transform fault links two other faults and accommodates the relative motion between the blocks of rock that move because of the displacement on the other two faults. However, we reserve the term <i>transfer fault</i> for this general type of displacement, independent of scale. |

**TABLE 8.2** — Types of Faults. Note the precise definitions of **Transform fault**, **Scissors fault**, and **Strike-slip fault**. The Tear fault definition is given in Chapter 18.

| TABLE 19.1         | TERMINOLOGY FOR STRIKE-SLIP FAULTS                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Flower structure   | An array of faults in a strike-slip fault zone that merges at depth into a near-vertical fault plane, but near the ground surface diverges so as to have shallower dips. In a positive flower structure, there is a component of thrusting on the faults, and in a negative flower structure, there is a component of normal faulting.                                                                                                                       |
| Horsetail          | A group of fault splays at the endpoint (fault tip) of a fault; the splays diverge to define a fanlike array in map view.                                                                                                                                                                                                                                                                                                                                    |
| Lateral escape     | The strike-slip movement of a crustal block in a direction perpendicular to the regional convergence direction in a collisional orogen; the block is essentially squeezed laterally out of the way of the colliding blocks.                                                                                                                                                                                                                                  |
| Lateral ramp       | A surface connecting two non-coplanar parts of a thrust fault or normal fault (i.e., a ramp that is roughly parallel to the transport direction). Strike-slip movement occurs on a lateral ramp.                                                                                                                                                                                                                                                             |
| Non-coplanar       | An adjective used to describe a geometric arrangement of two planes in which the planes have the same attitude, but are not aligned end-to-end to form a single plane.                                                                                                                                                                                                                                                                                       |
| Pull-apart basin   | A sedimentary basin that forms at a releasing bend along a strike-slip fault.                                                                                                                                                                                                                                                                                                                                                                                |
| Releasing bend     | In the context of describing a strike-slip fault, it is a map-view bend, in the fault plane, whose orientation is such that the block on one side of the fault pulls away from the block on the other, causing transtensional deformation.                                                                                                                                                                                                                   |
| Restraining bend   | In the context of describing a strike-slip fault, it is a map-view bend, in the fault plane, whose orientation is such that the block on one side of the fault pushes against the block on the other, causing transpressional deformation.                                                                                                                                                                                                                   |
| Stepover           | A geometry that occurs where the end of one fault trace overlaps the end of another, non-coplanar fault trace.                                                                                                                                                                                                                                                                                                                                               |
| Strike-slip duplex | An arrangement of sigmoidal-shaped fault splays (in map view) at a stepover, a restraining bend, or a releasing bend, whose geometry in map view resembles the cross-sectional geometry of the duplexes that occur in thrust-fault or normal-fault systems.                                                                                                                                                                                                  |
| Strike-slip fault  | Any fault on which displacement vectors are parallel to the strike of the fault, in present-day surface coordinates. The use of this term is purely geometric, and has no genetic, tectonic, or size connotation.                                                                                                                                                                                                                                            |
| Tear fault         | A traditional term for strike-slip faults that occur in a thrust sheet and accommodate differential displacement of one part of a thrust sheet relative to an adjacent part; in more recent jargon, a tear fault is a vertical lateral ramp.                                                                                                                                                                                                                 |
| Transcurrent fault | A strike-slip fault that has the following characteristics: it dies out along its length; the displacement across it is less than the length of the fault; the length of the fault increases with time and continued movement; displacement on the fault is greatest at the center of the fault trace and decreases toward the endpoints (tips) of the fault.                                                                                                |
| Transfer fault     | A fault trending roughly perpendicular to the axis of a rift that links together two non-coplanar parts of the rift; transfer faults accommodate map-view offset of the locus of extension. Kinematically, transfer faults are a type of transform fault. If a rift succeeds in evolving into a mid-ocean ridge, the transfer fault becomes an oceanic transform fault.                                                                                      |
| Transform fault    | A strike-slip fault that has the following characteristics: once formed, displacement across it can be constant along the length of the fault; displacement across it can be much greater than the length of the active fault; its length can be constant, increase, or decrease with time; it terminates at another fault. Transform faults can be lithosphere plate boundaries, in which case they terminate at intersections with other plate boundaries. |
| Transpression      | A combination of strike-slip and compressional deformation; transpression occurs where a fault is not parallel to the map projection of regional displacement vectors, so that there is a component of compression across the fault (leading to shortening).                                                                                                                                                                                                 |
| Transtension       | A combination of strike-slip and tension; transtension occurs where a fault is not parallel to the map projection of regional displacement vectors, so that there is a component of tension across the fault (leading to extension).                                                                                                                                                                                                                         |
| Wrench fault       | A traditional synonym for a strike-slip fault. It was commonly used in the petroleum geology literature, typically in reference to a regional-scale continental strike-slip fault. We abandon the term, as it tends to be used inconsistently.                                                                                                                                                                                                               |

**TABLE 19.1** — Complete terminology for strike-slip faults (Chapter 19). Note especially the definitions of **Transform fault**, **Tear fault**, **Transcurrent fault**, and **Wrench fault** — these directly answer why options (b), (c), and (d) are wrong.

SOURCE: VAN DER PLUIJM & MARSHAK • CHAPTER 19 • PAGE 478

## KEY FAULT TYPES — CONCEPT CARDS

### ✓ Transform Fault

**Plate boundary** strike-slip fault. Lithosphere neither created nor destroyed. Occurs in ocean basins linking mid-ocean ridge segments. Coined by J. Tuzo Wilson. Classic examples: Mid-Atlantic Ridge transforms, San Andreas (continental).

✗ **Tear Fault**

A strike-slip fault **within a thrust sheet** that accommodates differential displacement of one part of the sheet relative to another. Nearly vertical, parallel to transport direction. Found in thrust belts — NOT at plate boundaries.

✗ **Wrench Fault**

**Traditional synonym** for a continental strike-slip fault, used mainly in petroleum geology. Textbook explicitly states: "We abandon the term, as it tends to be used inconsistently." NOT an oceanic plate boundary feature.

✗ **Scissor (Scissors) Fault**

A fault where **slip changes along strike**, so the hanging wall block **rotates** around an axis perpendicular to the fault surface. A specific kinematic type — not a regional oceanic plate boundary.

COMPARISON TABLE — ALL FOUR FAULT TYPES

| FAULT TYPE               | NATURE OF SLIP                            | TECTONIC SETTING                                                        | PLATE BOUNDARY?                    | OCEAN FLOOR?                      |
|--------------------------|-------------------------------------------|-------------------------------------------------------------------------|------------------------------------|-----------------------------------|
| <b>Transform fault</b> ✓ | <b>Strike-slip; constant displacement</b> | <b>Links ridge segments, trench segments, or other plate boundaries</b> | <b>YES — conservative boundary</b> | <b>YES — most common in ocean</b> |
| Tear fault               | Dominantly strike-slip; vertical          | Within thrust sheets; accommodates differential displacement            | No                                 | No — thrust belt feature          |
| Wrench fault             | Strike-slip (continental)                 | Continental crust; petroleum geology term                               | Sometimes, but term abandoned      | No — continental usage            |
| Scissor fault            | Slip changes along strike; rotation       | Any setting with rotational block motion                                | No                                 | No — specific kinematic type      |

OPTION-BY-OPTION ANALYSIS

**a** ✓ **CORRECT ANSWER**

**"Transform faults"**

Directly defined by the textbook as plate boundaries where lithosphere is *neither created nor destroyed* (Table 8.2, p. 171). In ocean basins, they most commonly link segments of mid-ocean ridges, trending perpendicular to the ridge axis (§19.2.1, p. 479). These are the premier example of **regional oceanic strike-slip faults at lithospheric plate boundaries**. First described by J. Tuzo Wilson in the 1960s, their existence is one of the key proofs of plate tectonics.

**b** ✗ **INCORRECT**

**"Tear faults"**

Tear faults are **intra-thrust sheet** structures — they occur within thrust sheets to accommodate differences in how much one part of the sheet has moved versus another. Table 19.1 (p. 478) defines them as: "A traditional term for strike-slip faults that occur in a thrust sheet and accommodate differential displacement... in more recent jargon, a tear fault is a vertical lateral ramp." They are found in fold-thrust belts, not in the ocean at plate boundaries.

→ Correct setting for tear faults: **Fold-thrust belts** (e.g., lateral ramps in Appalachians, Himalayas)

**c** ✗ **INCORRECT**

## "Wrench faults"

Wrench fault is defined by Table 19.1 (p. 478) as simply: "A traditional synonym for a strike-slip fault. It was commonly used in the petroleum geology literature, typically in reference to a **regional-scale continental** strike-slip fault. We abandon the term, as it tends to be used inconsistently." While all transform faults are strike-slip faults, calling oceanic plate-boundary strike-slip faults "wrench faults" is imprecise and the term is explicitly abandoned by the textbook. The specific, correct, formally defined term for oceanic plate boundary strike-slip faults is *transform fault*.

→ Wrench fault = outdated/imprecise term for **continental** strike-slip faults

d

✗ INCORRECT

## "Scissor faults"

A scissor (scissors) fault is a specific kinematic type where **slip magnitude changes along strike**, causing one block to rotate around an axis perpendicular to the fault surface (Table 8.2, p. 171; Figure 8.4i). This is a purely geometric description — it says nothing about plate boundaries, ocean floors, or regional scale. Scissor faults are not a category of plate boundary.

→ Scissor fault = **rotational slip** type, not a plate boundary concept

### 🧠 EXAM MEMORY AID

**Transform fault** = Ocean + plate boundary + strike-slip → "*conservative boundary*"

**Tear fault** = Inside thrust sheets → differential displacement of sheet

**Wrench fault** = Abandoned term → used only for *continental* strike-slip

**Scissor fault** = Slip changes along strike → block *rotates*

*Key phrase from the question: "oceanic + plate boundary + regional" → only Transform fits*

### PROBABLE EXAM QUESTIONS FROM THIS TOPIC

#### PROBABLE Q · 1

**Transform faults are also called "conservative plate boundaries" because**

- (a) They conserve the kinetic energy of plate motion
- (b) They are ancient and geologically stable
- (c) Lithosphere is neither created nor destroyed along them
- (d) They conserve the orientation of the stress field

✓ **Answer: (c) — Van der Pluijm & Marshak, p. 357 (Ch.14) & Table 8.2, p. 171**

#### PROBABLE Q · 2

**The term "transform fault" was introduced by**

- (a) Harry Hess
- (b) Alfred Wegener
- (c) J. Tuzo Wilson
- (d) Lynn Sykes

✓ **Answer: (c) — Van der Pluijm & Marshak, §19.2.1, p. 479**

#### PROBABLE Q · 3

**Which of the following is NOT a characteristic of a transform fault?**

- (a) Displacement can exceed the length of the fault
- (b) It terminates at another fault or plate boundary
- (c) Displacement increases from center toward endpoints
- (d) Length can remain constant with time

✓ **Answer: (c) — That describes a transcurrent fault. Table 19.1, p. 478**

PROBABLE Q · 4

**A tear fault is characteristically found in**

- (a) Mid-ocean ridges linking two ridge segments
- (b) Transform plate boundaries in oceanic crust
- (c) Thrust sheets, accommodating differential displacement
- (d) Rift zones during early continental breakup

✓ **Answer: (c) — Table 19.1, p. 478 & §18.1, p. 447**

PROBABLE Q · 5

**Which of the following pairs represents the three types of plate boundaries?**

- (a) Normal, Reverse, Strike-slip
- (b) Divergent, Convergent, Transform
- (c) Rift, Subduction, Collision
- (d) Transcurrent, Transpressional, Transtensional

✓ **Answer: (b) — Van der Pluijm & Marshak, §14.8, p. 357 & Figure 14.14**

PROBABLE Q · 6

**On a scissors fault, the distinctive kinematic feature is that**

- (a) Strike-slip displacement is constant along the fault length
- (b) The hanging wall rotates around an axis perpendicular to the fault surface
- (c) It always occurs at mid-ocean ridges
- (d) Slip direction is parallel to the dip of the fault

✓ **Answer: (b) — Table 8.2, p. 171**